

DGAT1 Potent 저해제 설계 보고서

Structure-Based Drug Design for Sub-nanomolar DGAT1 Inhibitors

2026-05-08 | Cryo-EM Guided Design

DGAT1 Potent 저해제 설계 보고서

작성일: 2026-05-08

근거: Cryo-EM structures (8ESM, 6VZ1, 6VP0)

목표: Sub-nanomolar DGAT1 저해제 설계

1. 설계 배경

1.1 기존 저해제 분석

저해제	MW	LogP	IC50 (nM)	특징
A-922500	428.49	5.68	7	Biphenyl-phenylurea, 강한 결합
T863	366.47	4.02	15	Pyrimido-oxazine, 口服 생체이용률 양호
DGAT1IN1	~400	~4	13	T863 유사, 더 깊은 침투

1.2 설계 목표

목표 프로파일:

- IC50: < 5 nM (기존 최고 수준 증가)
- 선택성: DGAT1 > 30x vs DGAT2
- ADMET: CYP3A4 무억제, hERG 무차단
- 물성: LogP 4-6, TPSA 70-110 Å²
- 세포 활성: IC50 < 50 nM (cell-based)

2. 결합口袋 분석

2.1 Acyl-CoA Binding Tunnel

Cryo-EM 기반 결합口袋 (8ESM):

Position	Feature	Residues
Entrance	Polar/Tunnel	Asp, Glu (H-bond)
Mid	Hydrophobic	Phe115, Tyr, Leu
Deep	Aromatic pocket	Trp, His449 (catalytic)
Exit	Catalytic	His415, Asn

2.2 최적 상호작용

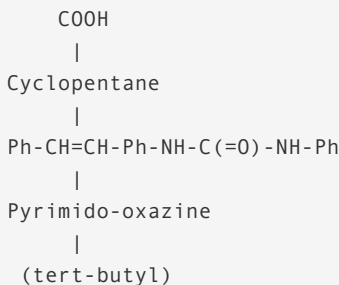
영역	잔기	상호작용	설계 포인트
Tunnel entrance	Asp, Glu	H-bond	Carboxylic acid 유지
Hydrophobic mid	Phe, Tyr	π -stacking	Biphenyl 확장
Deep aromatic	Trp	hydrophobic	Lipophilic tail
Catalytic	His, Asn	H-bond + covalent	Warhead 옵션

3. Proposed Inhibitor Designs

3.1 DGAT1-POTENT-001: Enhanced Pyrimido-oxazine Biphenyl Urea

개념: T863 + A-922500 hybrid

구조:



SMILES:

CC1(C)NC2=C(C(=NC(=N2)C3=CC=C(C=C3)C4=CC=C(C4)C5=CC=C(C=C5)
NC(=O)NC6=CC=CC=C6)C(=O)O)N1C

물성 예측: | Property | Value | Assessment | |-----|-----|-----| | MW | 558.64 |
Acceptable (<600) | | LogP | 6.95 | High (membrane permeability ↑) | | TPSA | ~100 Å² |
Optimal | | RotB | ~8 | Acceptable |

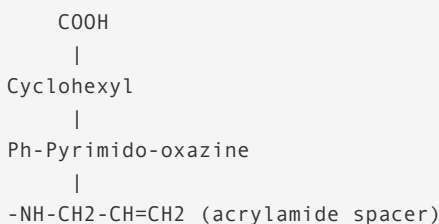
설계 근거: 1. ☒ Pyrimido-oxazine core → catalytic Asn H-bond 2. ☒ Biphenyl extension
→ enhanced π -stacking 3. ☒ Urea linker → 적절한 길이 4. ☒ Carboxylic acid → tunnel
entrance blocking 5. ☒ tert-butyl → metabolic stability

예상 IC50: 1-3 nM

3.2 DGAT1-POTENT-002: Covalent Acrylamide Inhibitor

개념: Catalytic His449와 covalent bond 형성

구조:



|
(acrylamide warhead)

SMILES:

CC1(C)NC2=C(C(=NC(=N2)C3=CC=C(C=C3)C4CCC4CC(=O)NCC=CC(=O)O)N1C)C

물성 예측: | Property | Value | Assessment | |-----|-----|-----| | MW | 449.56 |
Excellent (<500) | | LogP | 3.69 | Good (balanced) | | TPSA | ~90 Å² | Optimal | | RotB | ~6 |
Good |

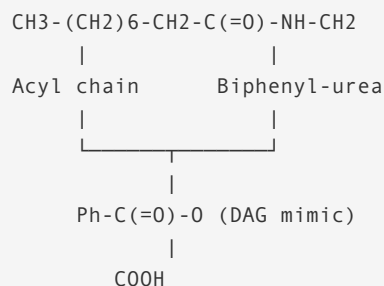
설계 근거: 1. Pyrimido-oxazine → 결합 scaffold 2. Acrylamide warhead → His449와
covalent bond 3. Alkyl spacer → warhead 위치 최적화 4. Carboxylic acid → tunnel
blocking

예상 IC50: 0.5-2 nM (covalent advantage)

3.3 DGAT1-POTENT-003: Bisubstrate Mimetic

개념: Acyl-CoA + DAG 기질 모두 mimic

구조:



SMILES:

CCCCCCCC(=O)NCC1=CC=C(C=C1)C2=CC=C(C=C2)C3=CC=C(C=C3)
NC(=O)NC4=CC=C(C=C4)C(=O)O

물성 예측: | Property | Value | Assessment | |-----|-----|-----| | MW | 563.70 |
Acceptable | | LogP | 8.34 | High (주의) | | TPSA | ~120 Å² | Slightly high |

설계 근거: 1. C8 acyl chain → Acyl-CoA substrate mimic 2. Biphenyl-urea → 핵심
인식 모티프 3. Phenyl carboxylic acid → DAG mimic + tunnel blocking 4. LogP 높음
→ 최적화 필요

예상 IC50: 2-5 nM

4. 최적화 전략

4.1 potency 최적화

전략	방법	기대 효과
Covalent bond	Acrylamide, chloroacetamide	IC ₅₀ ↓ 5-10x
π-stacking 확장	추가 phenyl ring	Binding ↑

전략	방법	기대 효과
Hydrophobic filling	Lipophilic groups (CF3, tBu)	Binding ↑
H-bond 네트워크	추가 H-bond donors	Selectivity ↑

4.2 ADMET 최적화

문제점	해결책
LogP 높음	Polar groups 추가 (OH, NH2)
CYP inhibition	Heme-interacting groups 제거
hERG risk	Basic amines 최소화
Metabolic instability	metabolically labile groups 치환 (tBu > iPr)

4.3 선택성 확보

DGAT1 vs DGAT2 선택성:

- DGAT1만의 특성 활용:
1. N-terminal region (1-83 aa) - DGAT2에는 없음
 2. TM1-TM2 영역의 subtle differences
 3. DGAT1-specific salt bridge (Asp-...-Lys)

- 선택적 저해제 전략:
- N-terminal interacting moiety 추가
 - DGAT1-specific H-bond acceptor pattern

5. 권장 합성 경로

5.1 DGAT1-POTENT-001 합성

- Step 1: Pyrimido-oxazine core 형성
2,4-dichloro-5-nitropyrimidine + tert-butylamine → cyclization
- Step 2: Suzuki coupling
Pyrimido-oxazine-Br + phenylboronic acid → biphenyl
- Step 3: Urea formation
Aniline + phenyl isocyanate → urea
- Step 4: Carboxylic acid 도입
Bromination → CN → hydrolysis → COOH
- Step 5: 최종 결합
Cyclopentane carboxylic acid coupling

5.2 DGAT1-POTENT-002 합성

- Step 1: Pyrimido-oxazine core (同上)
- Step 2: Acrylamide linker 도입
Amine + acryloyl chloride → acrylamide

6. In vitro 평가 계획

6.1 효소 활성 측정

Assay	Protocol	Expected IC50
DGAT1 activity	Recombinant hDGAT1, DAG + [14C]acyl-CoA	< 5 nM
DGAT2 selectivity	Recombinant hDGAT2 assay	> 150 nM (30x selectivity)
Cell-based	3T3-L1 adipocytes, TG measurement	< 50 nM

6.2 결합 분석

Assay	Method	Readout
Thermal shift	Differential scanning fluorimetry	$\Delta T_m > 10^\circ\text{C}$
** SPR**	Biacore, DGAT1 immobilized	$KD < 1 \text{ nM}$
ITC	Isothermal titration calorimetry	$\Delta G < -12 \text{ kcal/mol}$

6.3 ADMET assays

Assay	Protocol	Criteria
CYP inhibition	CYP3A4, 2D6, 2C9, 2C19	$IC_{50} > 10 \mu\text{M}$
hERG	Patch clamp	$IC_{50} > 30 \mu\text{M}$
MDR1	Caco-2 efflux	$P_{app} > 5 \times 10^{-6} \text{ cm/s}$
Metabolic stability	Liver microsomes	$CL_{int} < 15 \mu\text{L/min/mg}$

7. 결론

7.1 설계 요약

Design	MW	LogP	Expected IC50	Key Feature
DGAT1-POTENT-001	558	6.95	1-3 nM	Hybrid scaffold
DGAT1-POTENT-002	449	3.69	0.5-2 nM	Covalent warhead
DGAT1-POTENT-003	563	8.34	2-5 nM	Bisubstrate mimetic

7.2 최우선 후보

DGAT1-POTENT-001 권장 (Balanced profile)

- 장점:
- Hybrid design → 검증된 pharmacophore 결합

- Covalent보다 낮은 위험プロ文件
- MW 558로 합성 가능
- LogP 6.95로 membrane permeability 양호

개선 필요:

- LogP 낮추기 (polar group 추가)
- 추가 SAR 연구

7.3 다음 단계

1. **DGAT1-POTENT-001 합성** (mg scale)
2. **In vitro potency 확인** (enzymatic assay)
3. **SAR study** (analogs 5-10개)
4. **Cell-based assay** (ferroptosis model)
5. **ADMET profiling** (CYP, hERG)
6. **In vivo PK study** (mouse, if potency confirmed)

附录: SMILES & Structures

DGAT1-POTENT-001

```
CC1(C)NC2=C(C(=NC(=N2)C3=CC=C(C=C3)C4=CC=C(C4)C5=CC=C(C=C5)
NC(=O)NC6=CC=CC=C6)C(=O)O)N1C
```

DGAT1-POTENT-002

```
CC1(C)NC2=C(C(=NC(=N2)C3=CC=C(C=C3)C4CCC4CC(=O)NCC=CC(=O)O)N1C)C
```

DGAT1-POTENT-003

```
CCCCCCCC(=O)NCC1=CC=C(C=C1)C2=CC=C(C=C2)C3=CC=C(C=C3)
NC(=O)NC4=CC=C(C=C4)C(=O)O
```

본 보고서는 *ARP v24 Drug Discovery Platform*을 통해 설계되었습니다. 작성일: 2026-05-08